

Alam Bushara

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Education

University of Colorado - Boulder | Major GPA: 3.7+ | GPA: 3.5+

Boulder, CO

Bachelor of Science in Computer Science, Minor in Business Analytics.

Graduated: May 2025

Organizations: NSBE, Black Student Alliance, Sigma Nu, Theta Tau, Startup Summer

Relevant Coursework: Algorithms, Computer Systems, Cyber-Security, Data Structures, Database Systems, Human-Computer Interaction, Principles of Programming Languages, Software Development, Data Science, Theory of Computation, Object-Oriented Analysis, Numerical Computation, Artificial Intelligence.

Scholarships and Awards: Sachs Foundation Scholar, CU Esteemed Scholar, CU General Engineering Scholar

Skills

Programming Languages: Python, C++, JavaScript, Bash/Shell Scripting

Web & Backend: Node.js, React.js, FastAPI, TypeScript, Next.js, REST APIs, HTML, CSS

Data & Analytics: Palantir Foundry, Foundry AIP, Ontologies, Pipeline Builder, Data Modeling, Data Integration, ETL, SQL, Tableau, Power BI, Pandas, NumPy, scikit-learn, XGBoost

Databases: Microsoft SQL Server, MySQL, MongoDB, Oracle, IBI, PostgreSQL, SQLAlchemy, Alembic

Infrastructure: Git, Docker, Linux, CI/CD, Redis

Methods & Tools: Statistical Process Control (SPC), Root Cause & Corrective Action (RCCA), AS9100, Agile Development

Experience

Boeing Defense, Space, and Security

Huntsville, AL

Quality Data Engineer

November 2025 – Present

- Contributed to the integration of Palantir Foundry across quality engineering initiatives, developing Ontologies that unified engineering, manufacturing, and quality data to support the accelerated production ramp of advanced seeker systems.
- Developed SQL-based data pipelines and transformations using Pipeline Builder to integrate enterprise production and quality data, enabling automated reporting, non-conformance analysis, and reliability monitoring.
- Built operational applications and AI-powered workflows using Foundry AIP and LLM capabilities to automate quality investigations, accelerate root cause analysis, and improve engineering decision-making.
- Enforced data governance, role-based access controls, and security policies across sensitive engineering systems, ensuring audit compliance and integrity of production-critical data assets.

SB Energy

Denver, CO

Intern, Performance Engineering

May 2025 – October 2025

- Constructed Python analytics pipelines to monitor utility-scale solar performance, processing high-frequency time-series data to automate KPI reporting and identify degradation patterns across distributed plant systems.
- Built data ingestion and transformation systems to process large-scale drone-captured thermal and visual datasets, integrating outputs into Git-managed production workflows used by engineering and operations teams for real-time anomaly detection and output optimization.

National Security Agency

Boulder, CO

Intern, Software

August 2024 – May 2025

- Engineered an open-source toolkit leveraging Software-Defined Radio (SDR) for digital signal and protocol analysis, enhancing security assessments and RF threat detection within an Agile development environment.
 - Developed PyQt5-based real-time signal visualization tools featuring FFT analysis, multithreaded processing, and modular architecture while collaborating through Git-based version control and code reviews.
 - Reduced training time by 10%, saving the NSA over \$400,000 per recruit by optimizing core software workflows, improving onboarding efficiency, and streamlining engineering processes.
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Projects

Raumdeuter, World Cup 2026 Match Predictor

Tech Stack: Python, FastAPI, PostgreSQL, Redis, Next.js/TypeScript

- Designed a custom Elo rating engine (neutral-venue adjustment, recency decay, soft-ceiling compression) trained on 47K+ international matches, correcting for rating inflation in traditional Elo models.
- Architected a production backend (FastAPI, PostgreSQL, Redis caching, Alembic migrations, idempotent ETL) supporting 10K-simulation Monte Carlo tournament forecasts in under 2 seconds.
- Built a Next.js/TypeScript frontend with interactive pressing heatmaps and percentile-ranked player grades, powered by a possession-adjusted defensive metric engineered from StatsBomb tracking data.